

Replication README

This README describes the public replication package of *Legally Subjective* and is intended to be deposited alongside the manuscript wherever a venue asks for the materials needed to reproduce the analyses. The package lives at github.com/Vitalcheffe/legally-subjective and is organized so that a reviewer can walk the full chain — paper → code → corpus → frozen artifacts → predictions → regeneration — without credentials and without contacting the authors.

What regenerates from three commands

Stage	Script	What it reproduces
Corpus rebuild	<code>scripts/build_corpus.py</code>	Raw sources → frozen corpus with recomputed SHA-256 hashes
Baselines	<code>scripts/m2_baselines.py</code>	The pre-registered ideological rule and baseline accuracy figures
Challengers + statistics	<code>scripts/m3a_train.py</code>	Grouped cross-validation, McNemar tests, ablation — reproduces the challenger table exactly (all seeds printed)
Figures	<code>scripts/paper_figures.py</code> , <code>scripts/paper_figures_d1bis.py</code>	Every figure of the manuscript

Every CPU-side number in the paper regenerates from these scripts; randomness is seeded and seeds are printed in the outputs.

What is frozen on purpose

The GPU-dependent stages are frozen as executed artifacts rather than re-runnable scripts — the freeze is the point, because a sealed exam that can be re-taken is not sealed. The persona training (M3b) ran on free-tier GPU sessions under the pre-committed notebook `notebooks/m3b_qlora_personas.ipynb`, with machinery audited in `results/m3b_*`. The sealed exam (M4) was scored exactly once: predictions were generated by a kernel pinned to the `m4-freeze` head and frozen under a one-time seal whose lock (`results/m4_exam.lock`) records the head and the predictions' own SHA-256, alongside `results/m4_predictions.json` (commit `f6145a5`), before any scoring ran; `results/m4_report.json` is a deterministic function of those predictions and the frozen corpus. The mechanistic probes E1 and E3 ran under a kernel frozen before execution (tag `a1-freeze`), outputs under `results/a1_*.json`.

Leakage discipline (audited)

- Persona training pool: 477 rows — 0 sealed rows, 0 out-of-bounds rows (independent audit, re-executed by CI on every push).
- Sealed fifty selected by a deterministic seeded draw; seed derivation and resulting list hash are committed — regeneration-after-training is detectable from git history alone.
- Opinion-text corpus: 1,778 collected texts, normalized and deduplicated to 793 distinct documents with full provenance; fourteen-check zero-leak audit re-executed continuously.
- Sealed-chain hash check: all hashes consistent with `results/m4_exam.lock` at verification time.

How a reviewer verifies in five minutes

1. Clone the repository; confirm the three release tags exist (a1-freeze, m4-freeze, m4-exam).
2. Run scripts/build_corpus.py, scripts/m2_baselines.py, scripts/m3a_train.py in order; compare printed numbers to the manuscript tables.
3. Open results/m4_exam.lock; recompute the SHA-256 of results/m4_predictions.json and compare.
4. Inspect the zero-leak audit outputs under results/ and the CI history for the fourteen checks.
5. Optional: re-run scripts/m4_metrics.py to confirm the deterministic scoring of the frozen predictions.

Note for editors. Where a submission system asks for replication files as an upload, this README plus the repository URL satisfy the requirement without a bulky archive: the repository is the artifact, and it is public. A ZIP snapshot can be produced on request.